



THE COPPERBELT UNIVERSITY
SCHOOL OF GRADUATE STUDIES

CS 594: CLIMATE MODELLING AND DATA ANALYSIS

PRACTICAL 02:

LARS-WG 8 Multi-GCM Ensemble Analysis – Mount Makulu

Name: Fadzai Dzehonye

Student Number: 25184237

1. Task 1

The LARS-WG multi-GCM ensemble script was executed using the generated climate files for Mount Makulu. The analysis identified 21 .st files: one historical baseline and 20 future climate projections. The future projections represented five GCMs under SSP245 and SSP585 for the periods 2021–2040 and 2031–2050. The script then grouped the future files according to SSP and period and calculated ensemble climate statistics.

The *parse_st_filename()* function distinguishes the historical baseline from future projections using the filename. The file *mmkuWG.st* is explicitly identified as the baseline and assigned *is_baseline = TRUE*. For the remaining files, the function extracts the GCM, SSP and future period from the filename.

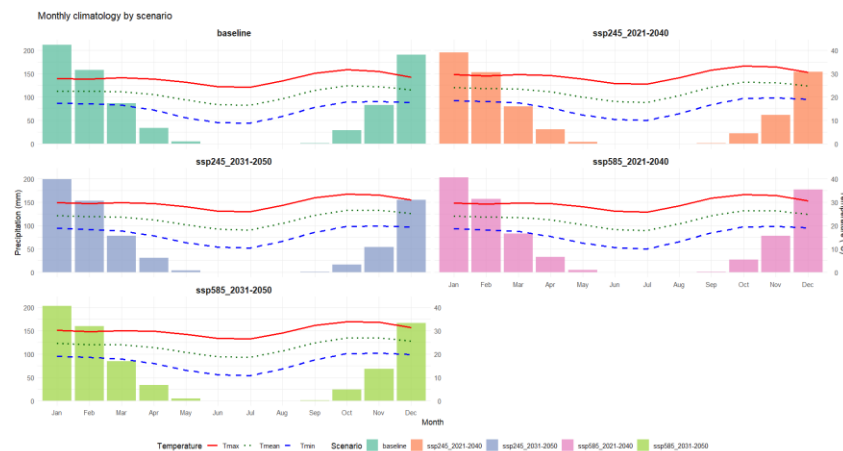


Figure 1: Monthly climatology of precipitation and temperature for the baseline and future SSP245 and SSP585 scenarios at Mount Makulu.

Figure 1 shows a strong seasonal rainfall pattern at Mount Makulu, with the highest precipitation occurring mainly between November and March and very low rainfall during the middle of the year. The future scenarios generally indicate higher maximum and minimum temperatures than the baseline, with greater warming evident in the later projection period (2031–2050).

Task 1 Reflection: The *parse_st_filename()* function distinguishes the baseline from future projections by identifying *mmkuWG.st* as the baseline file. For future files, it extracts the GCM name, SSP scenario and projection period from the filename. This enables the script to organise the generated files into the appropriate baseline and future scenario groups for ensemble analysis.

2. Task 2: Customising Visualisations and Seasons

The script was modified to use a custom colour palette and an alternative seasonal grouping. The original Brewer palette was replaced with custom colours, and the seasonal categories were changed from OND/JFM/AMJ/JAS to NDJ/FMA/MJJ/ASO.

```
# =====
# 2. User settings
# =====
WORK_DIR <- "C:/LARSWG8/Output/"
YEAR_OFFSET <- 1980
FILE_PATTERN <- "\\..st$"
OUTPUT_PREFIX <- "LARSWG_southern"

# Colour settings
PALETTE_TYPE <- "custom"          # "brewer" or "custom"
CUSTOM_COLOURS <- c(
  "#3B5B92",
  "#7FA6D8",
  "#8FB996",
  "#D9A066",
  "#B56576",
  "#6D597A"
)
BREWER_PALETTE <- "Set2"
..
```

The modified grouping represents NDJ, FMA, MJJ and ASO, while the custom palette improves visual distinction among the baseline and future climate scenarios.

```
# Southern Hemisphere seasons: NDJ, FMA, MJJ, ASO
assign_season <- function(month) {
  case_when(
    month %in% c(11, 12, 1) ~ "NDJ",
    month %in% c(2, 3, 4)   ~ "FMA",
    month %in% c(5, 6, 7)   ~ "MJJ",
    month %in% c(8, 9, 10)  ~ "ASO",
    TRUE ~ NA_character_
  )
}
```

3. Task 3

Seasonal climate anomalies were calculated by comparing each future scenario with the historical baseline using the formula:

$$\Delta X = X_{\text{future}} - X_{\text{baseline}}$$

where ΔX represents the climate anomaly. Positive temperature anomalies indicate warming relative to the baseline, while negative precipitation anomalies indicate reduced rainfall.

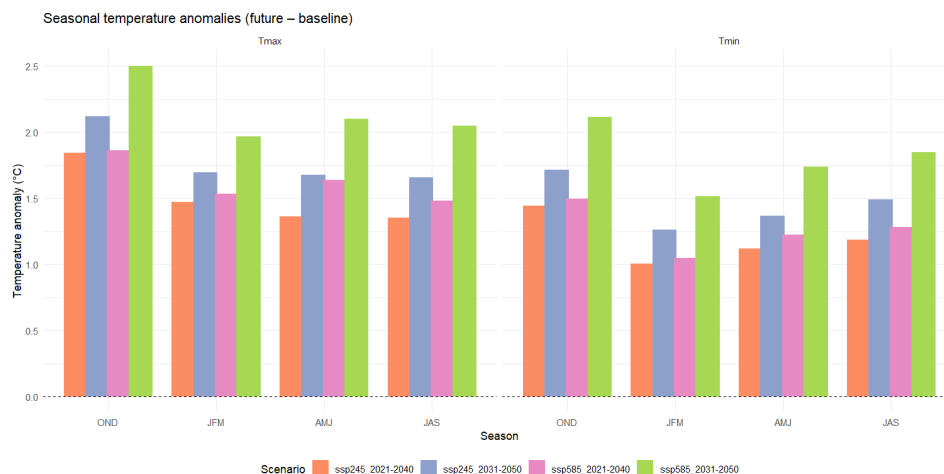


Figure 3: Seasonal maximum and minimum temperature anomalies for future climate scenarios relative to the historical baseline.

Task 3 Reflection: The greatest temperature increase in JFM is projected under SSP585 for 2031–2050, with T_{max} and T_{min} anomalies of approximately +1.97 °C and +1.52 °C, respectively. In JAS, precipitation decreases slightly under all future scenarios, with the largest reduction occurring under SSP245 for 2031–2050 (−0.642 mm).

Table 1: JFM Temperature and JAS Precipitation Anomalies Relative to the Historical Baseline

Future scenario	JFM Tmax anomaly (°C)	JFM Tmin anomaly (°C)	JAS precipitation anomaly (mm)
SSP245 2021–2040	+1.47	+1.00	−0.401
SSP245 2031–2050	+1.70	+1.26	−0.642
SSP585 2021–2040	+1.53	+1.05	−0.246
SSP585 2031–2050	+1.97	+1.52	−0.311

4. Task 4 Reflection:

LARS-WG is useful because it can generate site-specific daily climate series quickly and with relatively low computational requirements. It also allows efficient assessment of multiple GCMs and emission scenarios, which is helpful for climate impact and adaptation studies.

However, LARS-WG is a statistical weather generator and therefore depends on observed local climate characteristics and GCM-derived change factors. It may not fully represent complex atmospheric processes, spatial interactions or extreme events. Dynamically downscaled RCMs provide greater physical and spatial detail because they simulate regional climate processes directly, but they require much more computing power, storage and processing time.

5. Conclusion

The LARS-WG analysis demonstrated how historical and future climate data can be processed to assess projected changes in temperature and precipitation at Mount Makulu. The results indicate consistent future warming, with the strongest JFM warming projected under **SSP585 for 2031–2050**, while JAS precipitation shows small decreases across all scenarios. Overall, the exercise demonstrates the usefulness of multi-GCM ensemble analysis for understanding future climate variability and supporting climate adaptation and vulnerability assessments.